

The Watermark Theorem

The discriminant of the line lattice of Fermat surfaces of prime degree

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Abstract

Let $S_m \subset \mathbb{P}^3$ be the complex Fermat surface $x_0^m + x_1^m + x_2^m + x_3^m = 0$ of prime degree $m \geq 5$, and let V be the sublattice of $H^2(S_m, \mathbb{Z})$ generated by the classes of its $3m^2$ lines. We prove the closed formula

$$|\text{disc } V| = m^{3(m-3)^2}.$$

Since the lines generate the full Néron-Severi group for $\gcd(m, 6) = 1$ (Schütt-Shioda-van Luijk; Degtyarev), this computes $|\text{disc } \text{NS}(S_m)|$ exactly for every prime $m \geq 5$ and answers, for prime degree, the determinant question raised by Shioda in 1987 (Questions 7.2/7.4 of [7]). The proof is a chain of elementary, fully explicit lattice computations: a pairing lemma by character orthogonality, a trace-lattice discriminant by a Ramanujan-sum circulant, a transport identity, one exactly evaluated determinant, a blockwise saturation count, a kernel-saturation theorem ($mK \subseteq \text{SAT}$), and a 12-dimensional relation count over F_m solved in closed form. Every lemma was verified independently in a second computational environment, and the global identity was confirmed numerically at $m = 5, 7$ and, through a sealed pre-registered prediction, at $m = 11$. The companion note [B] derives the corollaries: the discriminant of the transcendental lattice and an explicit closed recipe for the Brauer group order of the reductions of S_m over finite fields.

1. Introduction

The Néron-Severi lattice of the Fermat surface has been a touchstone of the theory of algebraic cycles since Shioda's work of 1979–1987. For degree m with $\gcd(m, 6) = 1$, the $3m^2$ lines on S_m generate $\text{NS}(S_m)$ integrally — this is the theorem of Schütt-Shioda-van Luijk [4] completed by Degtyarev [5], resolving the question posed by Aoki-Shioda [3]. The *rank* of $\text{NS}(S_m)$ has been known in closed form since [3]:

$$\text{rank } \text{NS}(S_m) = 3(m-1)(m-2) + 1.$$

The *discriminant*, by contrast, resisted closure. Shioda asked for its structure in 1987 ([7], p. 133, Questions 7.2 and 7.4); Schütt-Shioda-van Luijk verified the value $|\text{disc } \text{NS}| = m^{3(m-3)^2}$ computationally for odd $m \leq 81$ ([4], around eq. (16)); Aljovin-Movasati-Villaflor [6] computed the elementary divisors of these lattices cell by cell for degrees up to 14 (their Table 1 — the same rows our instruments reproduce byte-exact in §13), at hardware costs reaching 170 GB of swap memory for a single cell, while their largest cells (the linear-cycle lattices behind their Table 2) remained beyond computation. To our knowledge no proof of the closed formula, valid for all prime m , existed in the literature: each published value is an isolated machine verdict.

This note proves it. The argument is deliberately elementary: it never leaves the explicit integer Gram matrix of the line configuration, and every step is either a character-orthogonality computation, a circulant determinant, or a finite-dimensional count over F_m carried out in closed form. The structure of the proof — discriminants of explicit trace sublattices, one big determinant, and the saturation index of the Gram kernel — was discovered by measurement (the campaign's instruments and run logs are part of the public record; see §13) and then closed by pen and paper. The same formula, with the extra factor $(m/4)^3$ for even m , fits all published data and seven sealed blind predictions of the campaign (the *Watermark Law*, [W]); only the odd-prime half is proven here, and the scope is stated precisely in §2.

2. The configuration, the lattice, and the statement

The surface and its lines. $S_m: x_0^m + x_1^m + x_2^m + x_3^m = 0$ in $P^3(C)$, $m \geq 5$ prime. S_m contains exactly $3m^2$ lines [4], organized in three families of m^2 according to the pairing of coordinates; in the standard parametrization (the one used in [4] and [6]) each family is indexed by $(k, l) \in (Z/m)^2$. By adjunction every line has self-intersection $2 - m$; two distinct lines of the same family meet (in one point) precisely when they share one of their two indices; lines of different families meet precisely when a single linear condition on the indices holds. In the coordinates fixed throughout this note (the campaign's session coordinates, which realize the standard configuration):

- within a family: $L_{\{(k,l)\}} \cdot L_{\{(k',l')\}} = 1$ if $k = k'$ or $l = l'$ (and $(k,l) \neq (k',l')$), 0 otherwise; self-intersection $2 - m$;
- family 1 $\times$ family 2: incidence $\iff k - l \equiv k' - l' \pmod{m}$;
- family 2 $\times$ family 3: incidence $\iff k + l \equiv k' + l' \pmod{m}$;
- family 1 $\times$ family 3: incidence $\iff k' - l' \equiv k + l + 1 \pmod{m}$.

Let $A = \bigoplus_{j=1,2,3} Z^{\{(Z/m)^2\}}$ be the free abelian group on the lines and G the $3m^2 \times 3m^2$ intersection Gram above. Define

$V := A / \ker G$, the *line lattice*, with the induced bilinear form.

All results below are theorems about this explicitly defined integer matrix; the geometric identification of G with the intersection form of the $3m^2$ lines is the classical configuration of [4], [6].

Theorem (the Watermark Theorem, odd-prime half). *For every prime $m \geq 5$,*

$$|\text{disc } V| = m^{\{3(m-3)^2\}}.$$

Corollary 1 (Shioda's question, prime degree). *For prime $m \geq 5$, $\gcd(m, 6) = 1$, so $V = NS(S_m)$ by [4, Thm 1.1] and [5]; hence $|\text{disc } NS(S_m)| = m^{\{3(m-3)^2\}}$, answering Questions 7.2/7.4 of [7] for prime degree.*

Scope. $m = 3$ is degenerate for this machinery (the four-pencil argument of Lemma R.2 and the degree count of Lemma C require $m \geq 5$) and its disc exponent is 0. Odd composite m and the even half of the Watermark Law (the extra factor $(m/4)^3$) are **not claimed here**: they remain conjectural, supported by the measured evidence catalogued in [W] (all twelve published AMV rows and seven sealed blind predictions).

Notation. For $t \in Z/m$ put $x_t(k, l) = k + t \cdot l$, and $x_\infty(k, l) = l$. The $m + 1$ **pencils** of a family are the level-set partitions of these forms ($t \in \{0, \infty\}$ the two **axis** pencils; the rest **live**). pb_L denotes pullback along x_L ; $Z_0 \subset Z^m$ the sum-zero sublattice; N_j the all-ones vector of family j ; $\text{aug}(u) = \sum u(p)$; $c_m(x) = m \cdot [x \equiv 0] - 1$ the prime- m Ramanujan function; $f_r := m \cdot e_r - 1 \in Z_0$, so $c_m(x - r) = f_r(x)$. The translation torus $\Gamma \cong (Z/m)^3$ acts on A by isometries; its characters act on the families through the dictionary $\text{fam1} \leftrightarrow (a, b, 0)$, $\text{fam2} \leftrightarrow (a, b, a+b)$, $\text{fam3} \leftrightarrow (a, a, a+b')$, and a family pencil corresponds to a punctured line of $(Z/m)^2$ -characters. The **trace vectors** are $v_{\{j,(a,b),r\}} :=$ the vector of family j with coordinates $c_m(ak + bl - r)$.

3. Lemma P (pairings, dead axes, phases, census)

Write $D_x := \sum_{k \in Z/m} \zeta^{\{xk\}} = m \cdot [x \equiv 0]$ and $w_j(a,b) := \sum_{k,l} \zeta^{\{ak+bl\}} L_j(k,l)$. Splitting each Gram block into its diagonal and incidence parts and summing geometric series gives closed pairing formulas; in particular:

(P1) For $a, b \neq 0$: $w_j(a,b) \cdot w_j(-a,-b) = (2-m)m^2 - m^2 - m^2 = -\mathbf{m}^3$, uniformly in m . **(P2)** For axis characters $((a,0)$ or $(0,b))$ the self-pairing vanishes and $w \perp H$ (the hyperplane class); negative-definiteness on the primitive part forces the axis classes to vanish in V — the *dead axes*. **(P-cross)** Blocks (1,2) and (2,3): $w_j(a,b) \cdot w_{\{j'\}}(a',b') = D_{\{a+a'\}} D_{\{a+b\}} D_{\{a'+b'\}}$. Block (1,3): the twisted incidence contributes one explicit phase: $w_1(a,b) \cdot w_3(a',b') = \zeta^{\{a'\}} \cdot D_{\{a+a'\}} D_{\{b+a'\}} D_{\{a'+b'\}}$. **(P3')** On the three overlap families of Γ -characters (where two family dictionaries collide) the supporting trace classes satisfy the null relations $v_{\{j,r\}} + v_{\{j', r+s_0\}} \in \ker G$, with phase law $s_0 = 0$ for the pairs (1,2), (2,3) and $s_0 = 1$ (the $\zeta^{\{a'\}}$ phase at function level) for (1,3). **(P4)** Inclusion-exclusion over the three blocks gives $3(m-1)^2 - 3(m-1) = \mathbf{3(m-1)(m-2)}$ live characters, each of multiplicity one in $V \otimes Q$ — reproving the rank formula of [3] from scratch.

Corollary 5 (kernel census). $\ker G \otimes Q =$ six axis character lines (rank $6(m-1)$) $\oplus$ three overlap null lines (rank $3(m-1)$) $\oplus$ a 2-dimensional trivial part; total rank $9m - 7$. In particular a kernel vector has zero isotypic mass on every live non-overlap pencil.

Verification: every displayed pairing checked exhaustively over all characters at $m = 5$ and 7 , including the $(1,3)$ phase on all m^4 cases per prime, in two independent environments.

4. Lemma T (the trace-lattice discriminant)

For a live orbit O supported on family j with pencil form (a', b') , the $m - 1$ independent trace vectors v_r have Gram $B(v_r, v_s) = \sum_t \zeta^{-t(r-s)} (w(\alpha) \cdot w(-\alpha)) = -m^3 \cdot c_m(r-s)$ by (P1); the $(m-1) \times (m-1)$ matrix $-m^3(mI - J)$ has $|\det| = m^{3(m-1)} \cdot m^{m-2}$, since $\det(mI - J)_{m-1} = m^{m-2}$. Hence for every live orbit

$$\text{disc } T(O) = \pm m^{4m-5}.$$

5. The transport identity

Let $K := \ker G$ (saturated, rank $9m - 7$) and let $F \subset A$ be the $\mathbb{Z}$ -span of one trace lattice per live orbit together with the trivial lift N_1 . Then $F \cap K = 0$ and

$$\text{unified} := [V : \varphi(F)] = [A : F \oplus K] = |\det[\text{basis of } K \mid \text{generators of } F]|,$$

where $\varphi: A \rightarrow V$ is the quotient map. Stacking the explicit kernel spawn K_0 (the Lemma-P basis: axis trace systems, paired overlap null systems, $N_1 - N_2$, $N_2 - N_3$) instead of a kernel basis measures $\text{unified} \cdot [K : K_0]$; the trivial lift satisfies $\varphi(N_1)^2 = m^3 = m^2 \cdot e^2$ where e generates the rank-1 trivial piece ($e^2 = m$), so $\varphi(N_1) = \pm m \cdot e$ contributes one exact factor m to the bookkeeping.

6. The big determinant

$$|\det[K_0 \mid F \mid N_1]| = m^{\{(9m^2-3m-6)/2\}}, \text{ exactly over } \mathbb{Z}.$$

Proof. Under the standard inner product on A : (i) different families have disjoint supports, hence are orthogonal; (ii) vectors built on different character lines within one family are orthogonal, because $\sum_{k,l} c_m(ak+bl-r) \cdot c_m(a'k+b'l-s) = (\sum_x c_m)(\sum_y c_m) = 0$ for independent forms — the map $(k,l) \mapsto (x,y)$ is a bijection of $(\mathbb{Z}/m)^2$ and Ramanujan sums are sum-zero. The standard Gram of the full stack is therefore block diagonal, and every block is a circulant of the pattern $\mu(m\delta - 1)$ with $\det(mI - J)_{m-1} = m^{m-2}$: six axis blocks and $3m - 9$ live non-overlap blocks contribute m^{3m-4} each (to $\det^2$); the three overlap blocks have pattern $m^2(mI - J) \otimes [[1,1], [1,2]]$ with $\det[[1,1], [1,2]] = 1$, contributing m^{6m-8} each; the trivial 3×3 block on $\{N_1 - N_2, N_2 - N_3, N_1\}$ has $\langle N_i, N_j \rangle = m^2 \delta_{ij}$, contributing m^6 . Total exponent $6(3m-4) + (3m-9)(3m-4) + 3(6m-8) + 6 = 9m^2 - 3m - 6$, halved for the determinant. Every block determinant is $\pm$ a power of m , so the identity is exact over $\mathbb{Z}$. ■

Verification: $v_m = 102$ ($m = 5$) and 207 ($m = 7$) with non- m part exactly 1 , both environments.

7. Blockwise saturation

Let $\text{SAT} \subset K$ be the explicit lattice: the six axis pullback blocks $\text{pb}_{\{j,\text{axis}\}}(Z_0)$; the three paired overlap blocks $\{\text{pb}_j(y) + \text{pb}_{\{j'\}}(y \circ \tau_{\{s_0\}}) : y \in Z_0\}$ (τ = cyclic translation, so the $P3'$ phase); and $Z(N_1 - N_2) \oplus Z(N_2 - N_3)$. Membership $\text{SAT} \subset K$ (which re-proves the phase law at the saturated level) and rank $9m - 7$ are file-verified. Each of the nine blocks saturates its trace sub-block by exactly m^{m-2} : the index $[Z_0 : \text{span}\{f_r\}]$ satisfies $\text{index}^2 = \text{disc}(\text{span } f_r) / \text{disc}(Z_0) = m^{2m-3}/m$, by the same circulant determinant; identically for the paired blocks, whose 2-power disc parts cancel in the ratio. Hence

$$[\text{SAT} : K_0] = m^{9(m-2)}.$$

8. Lemma R (mK ⊆ SAT)

R.1 (incidence identity). On each family, $\Sigma_{\{L \text{ over the } m+1 \text{ pencils}\}} E_L = m \cdot I + J$, where $(E_L u)(p) = \Sigma_{\{p' \in \text{fiber}_L(p)\}} u(p')$. *Proof:* a nonzero difference vector of $(\mathbb{Z}/m)^2$ lies in exactly one of the $m + 1$ directions; the diagonal is counted $m + 1$ times. ■

R.2 (augmentation divisibility). For $x = (x_1, x_2, x_3) \in K$: $\text{aug}(x_1) + \text{aug}(x_2) + \text{aug}(x_3) = 0$ exactly, and $m \mid \text{aug}(x_j)$ for each j . *Proof:* $G \cdot N_j = m \cdot 1_A$ by direct incidence counting, giving the exact sum on pairing with x . For the divisibilities, let y be the indicator of the family-1 fiber $\{k + t \cdot l \equiv c\}$, $t \neq 0$. Within family 1, $(Gy)_1 = 2 \cdot 1_1 - m \cdot y$ (for $t \neq 0$ the fiber is a graph over k and over l). Across families the incidence condition restricted to the fiber is a bijection except at one exceptional slope per block: $G \cdot y = 2 \cdot 1_1 + 1_2 + 1_3 - m \cdot y$ for $t \notin \{0, 1, -1\}$; $= 2 \cdot 1_1 + 1_2 + m \cdot z_3 - m \cdot y$ at $t = 1$; $= 2 \cdot 1_1 + m \cdot z_2 + 1_3 - m \cdot y$ at $t = -1$ (z the displaced fiber indicator in the exceptional family). Such t exists since $m \geq 5$. Pairing with x and writing $a_j = \text{aug}(x_j)$: $2a_1 + a_2 + a_3 \equiv 0$, $2a_1 + a_2 \equiv 0$, $2a_1 + a_3 \equiv 0 \pmod{m}$; subtracting, $a_2 \equiv a_3 \equiv 0$ and $2a_1 \equiv 0$ with m odd. ■ (All displayed identities verified entrywise for every t , c , both primes.)

R.3 (overlap assembly). For $x \in K$ and an overlap pair $(L \text{ in family } j) \leftrightarrow (L' \text{ in family } j')$ with shift s_0 : the sum-zero parts of the fiber-sum vectors satisfy $\tilde{f}_{\{j', L'\}} = \tilde{f}_{\{j, L\}} \circ \tau_{\{s_0\}}$. *Proof:* for pencils $L'' \neq L$ the map $p \mapsto (x_L(p), x_{L''}(p))$ is a bijection, so L -fibers kill L'' -isotypic mass; the trivial component contributes only the mean; hence $\tilde{f}_{\{j, L\}} = m \cdot g$ with $\text{pb}_L(g)$ the L -isotypic component of x_j . By the kernel census the only live pencils carrying kernel mass are the overlaps, where the component lies in the one-dimensional null direction of P_3' ; multiplying the a -th Fourier coefficient by $\zeta^{a \cdot s_0}$ is translation by s_0 , so $g' = g \circ \tau_{\{s_0\}}$. The pair $\text{pb}_j(\tilde{f}) + \text{pb}_{j'}(\tilde{f} \circ \tau)$ is then a paired-block member of SAT. ■

R.4 (theorem: mK ⊆ SAT). For $x \in K$, by R.1: $m \cdot x_j = \Sigma_L \text{pb}(f_{\{j, L\}}) - \text{aug}(x_j) \cdot 1_j$; the fiber means are integers by R.2; writing $f = \tilde{f} + \text{mean}$: $m \cdot x_j = \Sigma_L \text{pb}(\tilde{f}_{\{j, L\}}) + (\text{aug}(x_j)/m) \cdot 1_j$. Axis terms lie in the axis blocks of SAT; live non-overlap sum-zero parts vanish by the census; the six overlap terms group into the three R.3 pairs, members of SAT; and the constant leftover $\Sigma_j (\text{aug}_j/m) \cdot 1_j$ has integer coefficients summing to zero (R.2), hence lies in the span of $N_1 - N_3$, $N_2 - N_3$. So **mK ⊆ SAT**. ■ (Independent fire test: $m \cdot x \in \text{SAT}$ verified exactly for all 38 kernel basis vectors at $m = 5$.)

R.5. K/SAT is elementary abelian of exponent m ; $[K : \text{SAT}] = m^d$ with $d = (9m - 7) - \text{rank}_{\{F_m\}}(\text{SAT mod } m)$, and **the index has no part at any prime $p \neq m$** — by proof, not by measurement.

9. Lemma C (the counting lemma: d = 12)

$\text{rank}_{\{F_m\}}(\text{SAT mod } m) = 9m - 19$; equivalently $d = \dim_{\{F_m\}} K/\text{SAT} = 12$, for every prime $m \geq 5$.

Proof. Functions $F_m \rightarrow F_m$ are uniquely polynomials of degree $\leq m - 1$, and $\Sigma_{\{x \in F_m\}} x^e = -1$ if $(m-1) \mid e$, $e > 0$, and 0 otherwise; hence the mod- m span of the sum-zero column functions is exactly the polynomials of degree $\leq m - 2$. An F_m -relation among the SAT columns is a choice of polynomials $u_{\{j, k\}}$, $u_{\{j, l\}}$ (axis blocks), w_{12} , w_{23} , w_{13} (overlap blocks; one polynomial feeds both supporting families, the (1,3) block through the translation τ), and scalars ε_1 , ε_2 (the N -differences), satisfying three identities in $F_m[k, l]$ of degree $\leq m - 2 < m$ (so the function-polynomial dictionary is faithful):

$$(F1) \ u_{\{1, k\}}(k) + u_{\{1, l\}}(l) + w_{12}(k-l) + w_{13}(k+l) + \varepsilon_1 = 0 \quad (F2) \ u_{\{2, k\}}(k) + u_{\{2, l\}}(l) + w_{12}(k-l) + w_{23}(k+l) + (\varepsilon_2 - \varepsilon_1) = 0 \quad (F3) \ u_{\{3, k\}}(k) + u_{\{3, l\}}(l) + w_{23}(k+l) + (w_{13} \circ \tau)(k-l) - \varepsilon_2 = 0.$$

Mixed monomials $k^{d-r} l^r$ ($1 \leq r \leq d-1$) arise only from the two live-pencil terms; in degree d their coefficient is $C(d, r) \cdot [b_d + (-1)^r a_d]$, and $C(d, r) \neq 0 \pmod{m}$ for $d \leq m - 2$. For $d \geq 3$ the cases $r = 1, 2$ force $a_d = b_d = 0$; for $d = 2$ the single case $r = 1$ forces equality of the two quadratic coefficients, and chaining (F1)–(F3) (translation preserves the leading coefficient) makes all three equal to one common q — whose realizability is the polarization identity $(k-l)^2 + (k+l)^2 = 2k^2 + 2l^2$. Once the mixed terms cancel, the remaining content of each identity is k -pure plus l -pure, determining each family's axis pair up to its constant ambiguity (c , $-c$); the determined parts have degree $\leq 2 \leq m - 2$. The solution space is therefore parametrized freely by: the common quadratic q (1); the linear coefficients of the three w 's (3); their constants (3); ε_1 , ε_2 (2); the three constant splits (3) — total **12**, for every prime $m \geq 5$. ■

Verification: direct rank measurements 26 and 44 ($= 9m - 19$) by two independent constructions, plus the twelve relations exhibited explicitly (file COUNTING_LEMMA_PROBE_v1.py) and checked valid of rank 12 at both primes.

$$\text{Hence } [K : K_0] = [K : \text{SAT}] \cdot [\text{SAT} : K_0] = m^{\{12\}} \cdot m^{\{9(m-2)\}} = m^{\{9m-6\}}.$$

10. Assembly: proof of the Theorem

By §§5–9, $\text{unified} = |\det| / (m \cdot [K : K_0]) = m^{\{(9m^2-3m-6)/2 - 1 - (9m-6)\}} = m^{\{(9m^2-21m+4)/2\}}$. By §4 and Lemma T,

$$\text{disc } V = \text{disc}(\oplus_{\mathcal{O}} T(\mathcal{O}) \oplus Z \cdot e) / \text{unified}^2 = (m^{\{4m-5\}})^{\{3(m-2)\}} \cdot m / m^{\{9m^2-21m+4\}} = m^{\{(12m^2-39m+30) + 1 - (9m^2-21m+4)\}} = m^{\{3m^2-18m+27\}} = m^{\{3(m-3)^2\}}. \blacksquare$$

(The orbit count is $3(m-2) = 3(m-1)(m-2)/\varphi(m)$ for prime m .)

11. Sign and small cases

V contains the ample hyperplane class, so its form has hyperbolic signature $(1, \text{rank } V - 1)$ and $\text{disc } V = (-1)^{\{\text{rank } V - 1\}} |\text{disc } V|$ with $\text{rank } V = 3(m-1)(m-2) + 1$. $m = 3$: the exponent $3(m-3)^2$ is 0 and $\text{disc } V = \pm 1$ (classical; outside this proof's machinery). Odd composite and even degrees: conjectural — see [W].

12. Corollaries beyond the statement

- i. **Shioda's Questions 7.2/7.4** ([7], p. 133) are answered for prime degree (Corollary 1 above). (ii) **The transcendental lattice** $T_m = \text{NS}^\perp \subset H^2(S_m, \mathbb{Z})$ has $|\text{disc } T_m| = m^{\{3(m-3)^2\}}$ and $\text{rank } (m-1)(m-3)^2$ — see the companion note [B], where this is derived from unimodularity of H^2 in five lines. (iii) **The Brauer group** of the reductions of S_m over finite fields acquires a closed-form order recipe through the Artin-Tate formula, every ingredient of which is now explicit — see [B]. (iv) The glue group $V/(\oplus \text{ pieces})$ is proven elementary abelian of F_m -dimension $(3m^2-9m+4)/2$, a structure result on the integral cohomology of independent interest.

13. Computational verification and provenance

Every measured number in this note reproduces from instruments in the public repository: GRANSLAP_PROBE_v1/v2 (the big determinant, the independent full-kernel index, the SNF of G — whose nonzero elementary-divisor profile coincides with the published AMV rows), LEMAR_SHAKE_PROBE_v1 (residual anatomy at both primes), BETA1_DISPLAY_PROBE_v1 (the R.2 display identities for every parameter; the three pure generators), COUNTING_LEMMA_PROBE_v1 (the rank and the twelve explicit relations). The development followed a dual-instance protocol: every lemma was independently re-derived, and every computation independently reproduced, in a second environment before ratification; two intermediate conjectures were killed by their own pre-registered tests (the failures are documented in the campaign findings), and the $m = 11$ cell was decided by a sealed prediction written before computation — 27 orbit pieces, each of discriminant 11^{17} , glue 11^{134} — which passed on all counts (engine LACAZADELOSPÉLADOS, run log in repository; MacBook Air M2, 8 GB, single thread, 25% CPU). The final identities of §§6–9 imply, for the $m = 11$ rail, big determinant 11^{525} , $[K : K_0] = 11^{93}$, unified 11^{431} , all consistent with the sealed run.

References

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[W] *The Watermark Law* (this repository: THE_WATERMARK_LAW.md) — the full empirical law, the twelve AMV rows, the seven sealed blind predictions, and the published off-by-one erratum. [B] *Corollaries of the Watermark Theorem: the transcendental lattice and the Brauer group* (this repository: THE_BRAUER_COROLLARY.md).

Numbers from files only. Every claim above is either proven in the text or anchored to a named instrument and run log in the repository.